# U23CS704

# CRYPTOGRAPHY AND NETWORK SECURITY

**Proposed Sample Practice Assessment**

| Time (assumed) | Maximum marks (assumed) | Assessment status |
|---|---|---|
| 2 hours | 50 | Practice paper for capstone illustration only; not an official university examination pattern. |

Instructions: Answer all questions. Show all steps for numerical questions, state assumptions, and use clear protocol diagrams where useful. This sample has equal unit marks (10 per unit). The uploaded syllabus confirms CIE = 40 and SEE = 60 but does not specify examination duration or question pattern.

## Section A — Short Answers (5 × 2 = 10 marks)

- Q1. [2 marks — K2 — CO1 — Unit I] Distinguish a security attack from a security service. Give one example of each.
- Q2. [2 marks — K3 — CO2 — Unit II] In RSA, let p = 11 and q = 13. Compute n and φ(n). (No need to compute a private key.)
- Q3. [2 marks — K2 — CO3 — Unit III] State two differences between a cryptographic hash and a digital signature.
- Q4. [2 marks — K2 — CO4 — Unit IV] What does HTTPS add to ordinary HTTP, and why is certificate validation important?
- Q5. [2 marks — K2 — CO5 — Unit V] State the primary purpose of IPsec Authentication Header (AH) and Encapsulating Security Payload (ESP).

## Section B — Application and Design (5 × 8 = 40 marks)

- Q6. [8 marks — K4 — CO1/CO2 — Unit I] A college portal is accessed from public Wi-Fi. Identify four plausible threats, map each threat to a security objective, and recommend a suitable security service or mechanism. Explain why each recommendation is relevant.
- Q7. [8 marks — K4 — CO2/CO4 — Unit II] Compare AES and RSA with respect to key type, typical role, performance and key distribution. For a file-sharing system, recommend how the two could be used together and justify your design.
- Q8. [8 marks — K5 — CO3 — Unit III] A department must distribute signed notices and detect message alteration. Evaluate SHA-512, HMAC/CMAC and digital signatures for this case. Propose key management and public-key validation using Kerberos or X.509 concepts where appropriate.
- Q9. [8 marks — K4 — CO4/CO5 — Unit IV] Explain the main steps in establishing a secure HTTPS session and describe what SSH is used for. Analyse the risk of ignoring a browser certificate warning and recommend a safer response. Add two protections for a campus wireless LAN.
- Q10. [8 marks — K6 — CO3/CO5 — Unit V] Design an e-mail and network-layer security plan for confidential examination material communicated between campuses. Use relevant PGP or S/MIME concepts and IPsec architecture including AH/ESP and security associations. State assumptions and explain the controls chosen.

## Paper design tags

| Question | Unit | CO | Marks | Draft Bloom |
|---|---|---|---|---|
| Q1 | I | CO1 | 2 | K2 |
| Q2 | II | CO2 | 2 | K3 |
| Q3 | III | CO3 | 2 | K2 |
| Q4 | IV | CO4 | 2 | K2 |
| Q5 | V | CO5 | 2 | K2 |
| Q6 | I | CO1/CO2 | 8 | K4 |

| Question | Unit | CO | Marks | Draft Bloom |
|---|---|---|---|---|
| Q7 | II | CO2/CO4 | 8 | K4 |
| Q8 | III | CO3 | 8 | K5 |
| Q9 | IV | CO4/CO5 | 8 | K4 |
| Q10 | V | CO3/CO5 | 8 | K6 |
| Total | I–V | All COs | 50 | Mix |

Audit notice: this is a newly authored sample paper for the capstone. The named blooms-quality-and-rebalance-report skill cannot generate the formal original-question-paper audit until the actual CNS question paper is attached. Once attached, the original should be audited on all seven criteria and only items below the skill's weak threshold regenerated.